Block practice · Further cells, reproduction and disease

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Describe the steps of IVF treatment

Make alginate beads and give uses and advantages of immobilised enzymes

Relate sperm structures to their roles in fertilisation

State the symptoms and effects of inherited conditions

Explain oral rehydration and treatment of diarrhoeal disease

Explain sewage treatment, pit latrine design and decomposer roles

Describe virus structure and reproduction

Explain genetic and environmental causes of variation

Interpret human growth data, proportions and percentile charts

Explain how the placenta, cord and hormones support a fetus

Interpret antibacterial culture investigations

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Practice workbook · Describe the steps of IVF treatment

In-vitro fertilisation from hormone stimulation through egg collection to embryo transfer.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DESCRIBE-IVF-PROCEDURE

1. Question 6 · part c · 4 marks

June 2024 Paper 1R

Worked-example retry · Attempt part c only.

6 (a) The table gives the names and functions of hormones in the menstrual cycle.

Complete the table by placing one tick (✓) in each row of the table to show the main function of each hormone in the menstrual cycle.

(3)

Hormone	Function of hormone			
	causes ovulation	builds up uterus lining after menstruation	maintains uterus lining during pregnancy	stimulates a follicle to mature
progesterone				
FSH				
oestrogen				

(b) The chance of pregnancy can be decreased by using contraception.

State **one** method of barrier contraception.

(1)

(c) In-vitro fertilisation (IVF) can be used to increase the chances of pregnancy.

Describe the process of IVF.

(4)

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1. Question 6 · part c · 4 marks

June 2024 Paper 1R

Worked-example retry · Attempt part c only.

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(d) Many mothers decide to breastfeed their new-born baby.

Describe the advantages of breastfeeding.

(2)

(Total for Question 6 = 10 marks)

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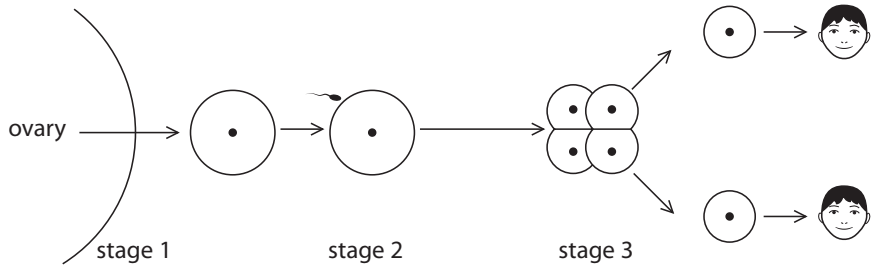
Working and answer

2. Question 5 · part b.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.i only.

5 (a) The diagram shows how genetically identical twins are produced.



(i) Name the process that occurs at stage 1.

(1)

(ii) Name the process that occurs at stage 2.

(1)

(iii) Use information from the diagram to explain how genetically identical twins are formed.

(2)

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2. Question 5 · part b.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.i only.

(b) (i) In-vitro fertilisation (IVF) increases the chance of producing twins.

Which hormones are given to a female at the start of IVF treatment?

(1)

- ☒ **A** FSH and LH
- ☒ **B** LH and oestrogen
- ☒ **C** oestrogen and progesterone
- ☒ **D** progesterone and FSH

(ii) Describe the procedures in IVF after hormone treatment.

(4)

(Total for Question 5 = 9 marks)

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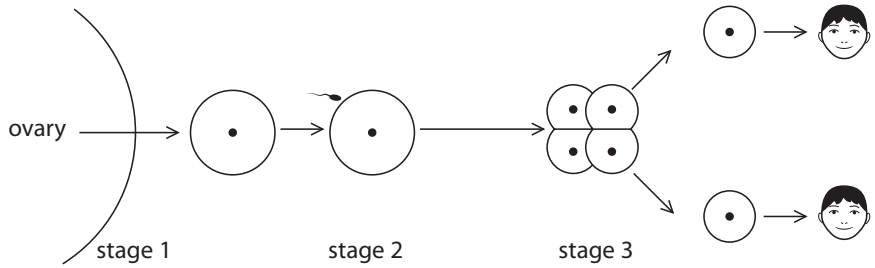
Working and answer

3. Question 5 · part b.ii · 4 marks

June 2019 Paper 2

Further practice · Attempt part b.ii only.

5 (a) The diagram shows how genetically identical twins are produced.



(i) Name the process that occurs at stage 1.

(1)

(ii) Name the process that occurs at stage 2.

(1)

(iii) Use information from the diagram to explain how genetically identical twins are formed.

(2)

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3. Question 5 · part b.ii · 4 marks

June 2019 Paper 2

Further practice · Attempt part b.ii only.

(b) (i) In-vitro fertilisation (IVF) increases the chance of producing twins.

Which hormones are given to a female at the start of IVF treatment?

(1)

- ☒ **A** FSH and LH
- ☒ **B** LH and oestrogen
- ☒ **C** oestrogen and progesterone
- ☒ **D** progesterone and FSH

(ii) Describe the procedures in IVF after hormone treatment.

(4)

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(Total for Question 5 = 9 marks)

3. Question 5 · part b.ii · 4 marks · Working and answer

Working and answer

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4. Question 8 · part a.ii · 3 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

8 (a) When women cannot conceive naturally, they can have fertility treatment.

Women having fertility treatment can be given the hormones FSH and LH.

(i) Give a role of LH in women.

(1)

(ii) Giving birth to twins or triplets is described as a multiple birth.

Explain how the use of FSH to treat infertility could increase the chances of a multiple birth.

(3)

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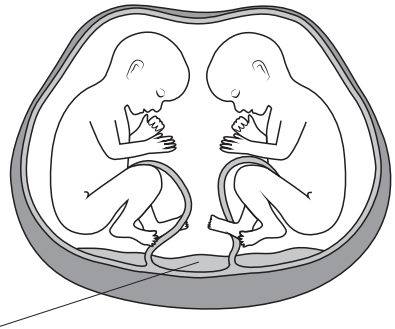
OLevels.COM SCHOOL

4. Question 8 · part a.ii · 3 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

(b) Some identical twins share a placenta, as shown in the diagram.



(i) Twins that share a placenta are often born with a lower mass than twins that have separate placentas.

Explain why twins sharing a placenta have a lower mass when they are born.

(3)

(ii) Identical twins form when the embryo splits early in pregnancy.

Explain why the twins are genetically identical.

(2)

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4. Question 8 · part a.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Make alginate beads and give uses and advantages of immobilised enzymes

Enzymes fixed in alginate beads for lactose-free milk or sugar conversion: preparation, the enzyme named and why immobilising helps.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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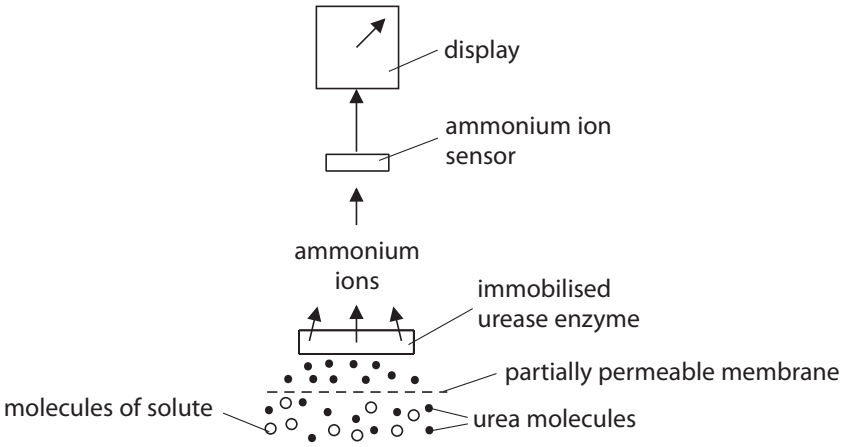
Type ID: HBI-IMMOBILISED-ENZYMES

1. Question 3 · part c · 2 marks

June 2021 Paper 2

Worked-example retry · Attempt part c only.

3 The diagram shows a biosensor that can be used to measure the concentration of urea in urine.



(a) (i) The urea molecules are in solution.

State the name of the solvent. **(1)**

(ii) Explain the function of the partially-permeable membrane. **(2)**

(iii) Suggest the function of the urease enzyme. **(3)**

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P 6 6 4 3 3 A 0 6 2 4

1. Question 3 · part c · 2 marks

June 2021 Paper 2

Worked-example retry · Attempt part c only.

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(b) A scientist wants to use the biosensor to compare the urea concentration of two urine samples.

Explain why it is important to do the tests at the same temperature.

(4)

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(c) Immobilised enzymes are also used in industry to produce useful molecules.

Give two advantages of using immobilised enzymes instead of enzymes that are free to move in solution.

(2)

1 .....

2 .....

(Total for Question 3 = 12 marks)

1. Question 3 · part c · 2 marks · Working and answer

Working and answer

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2. Question 6 · part a.i · 1 mark

November 2020 Paper 1

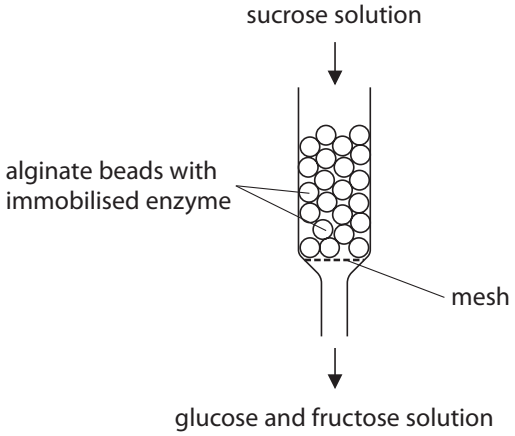
Further practice · Attempt part a.i only.

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6 (a) The diagram shows how immobilised enzymes can be used to convert sucrose into glucose and fructose.



(i) Name the immobilised enzyme used in this process.

(1)

.....

(ii) One advantage of using immobilised enzymes is that they do not contaminate the products, glucose and fructose.

Explain how the products could be tested to show that they had not been contaminated by the enzyme.

(3)

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(iii) State two other advantages of using immobilised enzymes.

(2)

1 .....

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2 .....

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2. Question 6 · part a.i · 1 mark · Working and answer

Working and answer

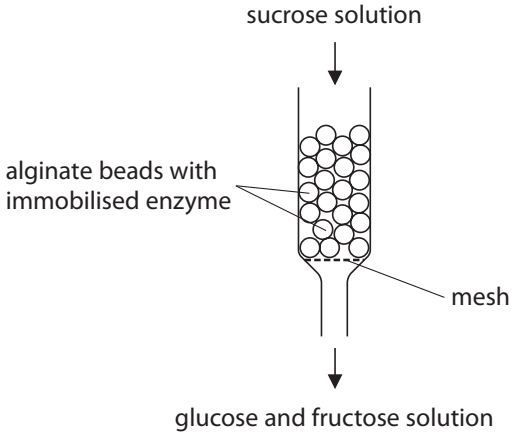
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3. Question 6 · part a.iii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.iii only.

6 (a) The diagram shows how immobilised enzymes can be used to convert sucrose into glucose and fructose.



(i) Name the immobilised enzyme used in this process.

(1)

.....

(ii) One advantage of using immobilised enzymes is that they do not contaminate the products, glucose and fructose.

Explain how the products could be tested to show that they had not been contaminated by the enzyme.

(3)

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(iii) State two other advantages of using immobilised enzymes.

(2)

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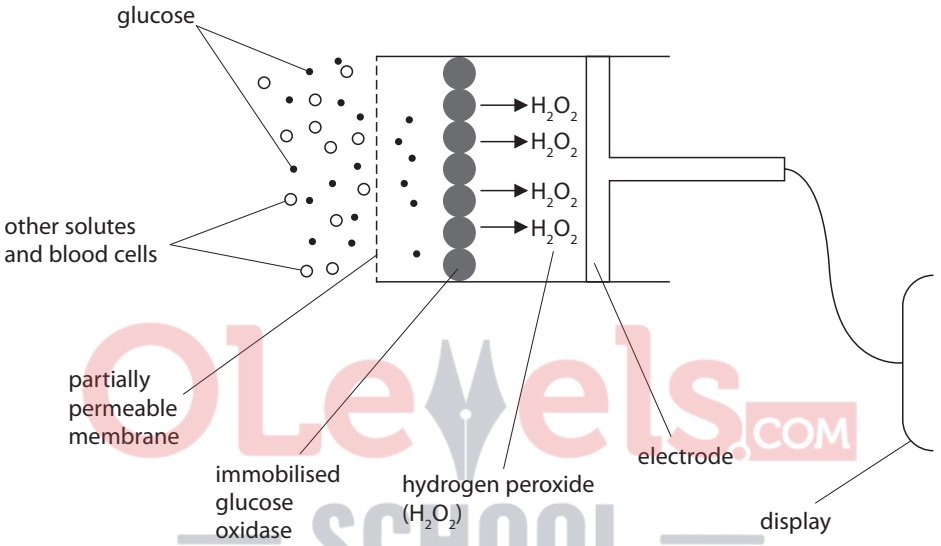
3. Question 6 · part a.iii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.iii only.

(b) Biosensors that contain immobilised enzymes can be used to measure the levels of glucose in blood.

The diagram shows this type of biosensor.



(i) Explain the function of the partially permeable membrane.

(2)

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3. Question 6 · part a.iii · 2 marks · Working and answer

Working and answer

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4. Question 7 · part a.i · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.i only.

7 (a) Industrial processes can use immobilised enzymes.

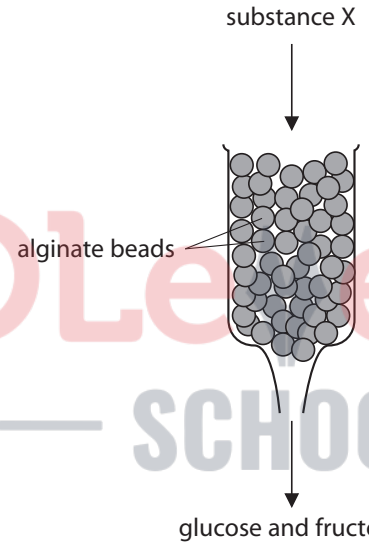
(i) Give one reason why using immobilised enzymes can cost less money than using enzymes that are not immobilised.

(1)

.....

.....

(ii) The diagram shows the apparatus used to produce glucose and fructose using immobilised enzymes attached to alginate beads.



Name substance X.

(1)

.....

(iii) Describe how to prepare alginate beads.

(3)

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4. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 7 · part a.ii · 1 mark

June 2022 Paper 1

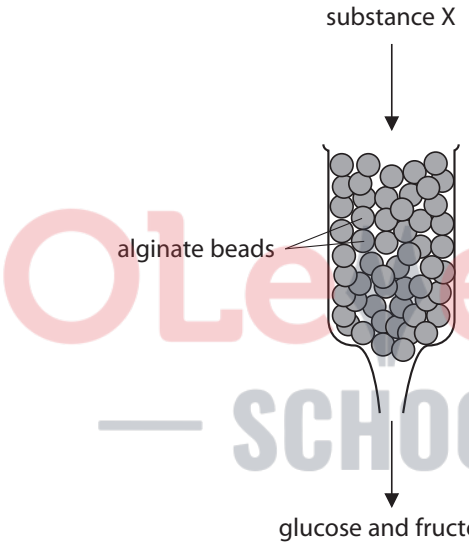
Further practice · Attempt part a.ii only.

7 (a) Industrial processes can use immobilised enzymes.

(i) Give one reason why using immobilised enzymes can cost less money than using enzymes that are not immobilised.

(1)

(ii) The diagram shows the apparatus used to produce glucose and fructose using immobilised enzymes attached to alginate beads.



Name substance X.

(1)

(iii) Describe how to prepare alginate beads.

(3)

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5. Question 7 · part a.ii · 1 mark · Working and answer

Working and answer

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6. Question 6 · part c.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part c.i only.

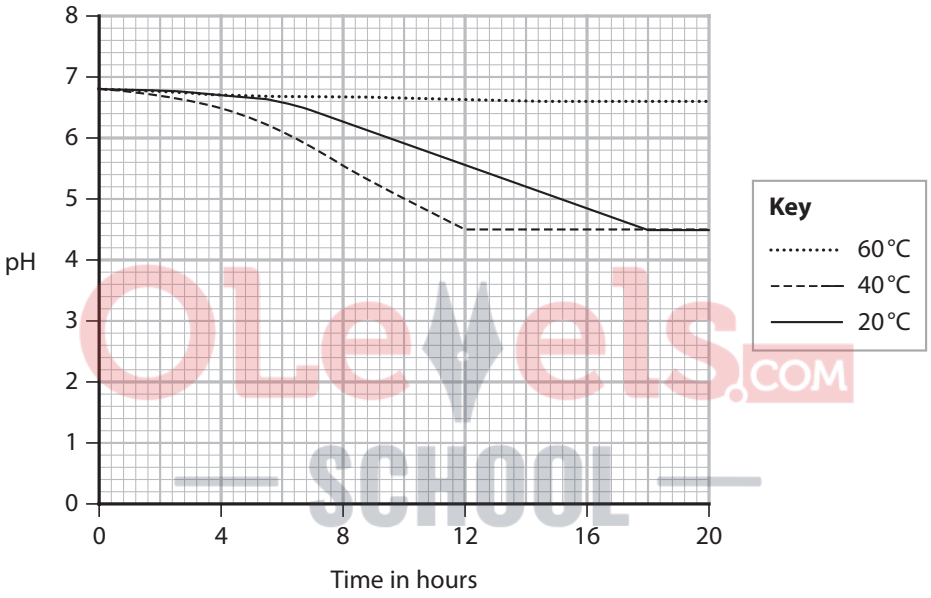
6 Yoghurt can be produced by incubating milk with a small amount of live yoghurt.

The live yoghurt contains *Lactobacillus* bacteria.

Enzymes convert lactose in the milk into lactic acid causing the pH to decrease to around pH 4.4.

In an investigation, the change in pH as yoghurt is produced was measured during 20 hours at three different temperatures.

The graph shows the results.



(a) (i) Calculate the difference in the time taken to produce yoghurt at 40 °C instead of 20 °C.

(2)

time = ... hours

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6. Question 6 · part c.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part c.i only.

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(ii) Give two variables that need to be controlled in this investigation.

(2)

1 .....

.....

2 .....

.....

(iii) Describe how this investigation could be improved to find the optimum temperature for the production of yoghurt.

(2)

.....

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(iv) In some methods to produce yoghurt, the milk is boiled and then cooled before the bacteria are added.

Give the reason for boiling and the reason for cooling the milk.

(2)

boiling

.....

.....

cooling

.....

.....

6. Question 6 · part c.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part c.i only.

(b) When yoghurt reaches a pH of 4.4 the bacteria stop fermenting lactose.

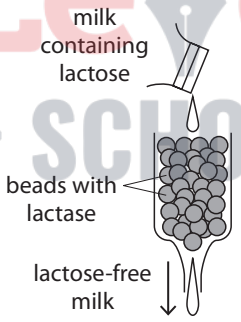
Explain why bacteria stop fermenting the lactose at a pH of around 4.4

(2)

(c) People who are lactose intolerant can drink lactose-free milk.

This can be produced using alginate beads containing lactase.

The diagram shows the equipment needed.



(i) Describe the advantages of using alginate beads containing lactase rather than a solution of lactase.

(2)

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6. Question 6 · part c.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part c.i only.

(ii) The enzyme lactase breaks down lactose into simpler sugars.

Explain how the structure of the small intestine allows the absorption of these sugars.

(3)

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(Total for Question 6 = 15 marks)

Working and answer

7. Question 7 · part a.iii · 3 marks

June 2022 Paper 1

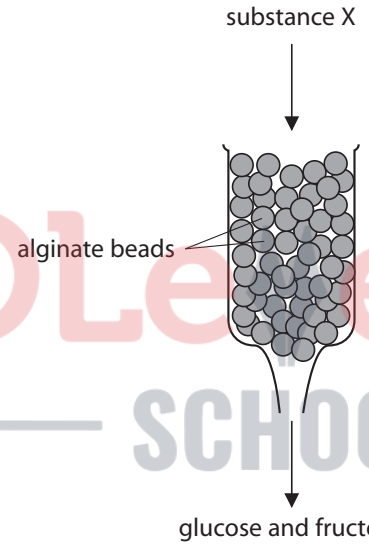
Further practice · Attempt part a.iii only.

7 (a) Industrial processes can use immobilised enzymes.

(i) Give one reason why using immobilised enzymes can cost less money than using enzymes that are not immobilised.

(1)

(ii) The diagram shows the apparatus used to produce glucose and fructose using immobilised enzymes attached to alginate beads.



Name substance X.

(1)

(iii) Describe how to prepare alginate beads.

(3)

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7. Question 7 · part a.iii · 3 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

(b) Which process is involved in the conversion of glucose into glycogen in the human body?

(1)

- ☒ **A** glucagon is released from the pancreas and travels to the liver
- ☒ **B** insulin is released from the pancreas and travels to the liver
- ☒ **C** glucagon is released from the liver and travels to the pancreas
- ☒ **D** insulin is released from the liver and travels to the pancreas

(c) (i) The total population of the United Kingdom is 66.65 million.

Diabetes affects 6.0% of this population.

Calculate the number of people with diabetes in the United Kingdom.

(2)

number of people = ... million

(ii) Surveys were carried out in different areas of the United Kingdom.

These surveys collected data on the number of people in each area with diabetes.

The information collected is shown in the box.

38	62	51	96	49	78	82	91	65

Determine the median number of people with diabetes.

(2)

median = ...

(Total for Question 7 = 10 marks)

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7. Question 7 · part a.iii · 3 marks · Working and answer

Working and answer

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Practice workbook · Relate sperm structures to their roles in fertilisation

Labelling sperm parts and explaining the roles of the tail, mitochondria-packed midpiece, acrosome and haploid nucleus.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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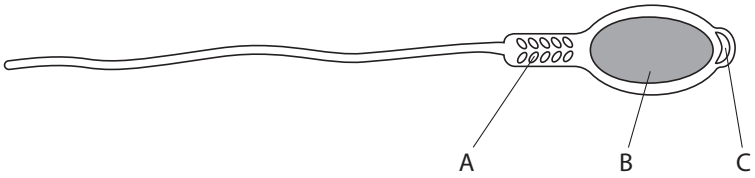
Type ID: HBI-RELATE-SPERM-STRUCTURE-FUNCTION

1. Question 6 · part a.ii · 3 marks

November 2023 Paper 2

Worked-example retry · Attempt part a.ii only.

6 The diagram shows a human sperm.



(a) (i) Name the parts A, B and C.

(3)

A .....

B .....

C .....

(ii) Describe the function of part C.

(3)

.....

.....

.....

.....

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.....

(iii) The actual length of the sperm is 50.0 micrometres (μm).

Calculate the magnification of the diagram.

[1 mm = 1000 μm]

(4)

magnification = ...

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1. Question 6 · part a.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 5 · part a.i · 2 marks

June 2021 Paper 1

Further practice · Attempt part a.i only.

5 The diagram shows a sperm.



(a) (i) Describe the function of a sperm.

(2)

(ii) State two other structures, apart from mitochondria, that can be seen in the sperm if it is viewed using an electron microscope.

(2)

1 .....

2 .....

(b) Explain why there are a large number of mitochondria found in a sperm.

(3)

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2. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 5 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

5 The diagram shows a sperm.



(a) (i) Describe the function of a sperm.

(2)

(ii) State two other structures, apart from mitochondria, that can be seen in the sperm if it is viewed using an electron microscope.

(2)

1 .....

2 .....

(b) Explain why there are a large number of mitochondria found in a sperm.

(3)

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3. Question 5 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

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(c) The actual length of the sperm between X and Y on the diagram is 3 µm.

Calculate the magnification of the sperm shown in the diagram.

[1000 µm = 1 mm]

(4)

magnification = × .....

(Total for Question 5 = 11 marks)

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Working and answer

4. Question 2 · part a.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.i only.

2 (a) A human sperm cell has a tail, a mid-piece containing organelles and a head with a nucleus.

(i) Give the function of the tail.

(1)

.....

.....

(ii) Which of these, found in the mid-piece, release energy?

(1)

- ☒ **A** chromosomes
- ☒ **B** endoplasmic reticulum
- ☒ **C** mitochondria
- ☒ **D** ribosomes

(iii) The process of meiosis is involved in the production of sperm.

Explain the importance of meiosis in sperm production.

(3)

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4. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 2 · part b.ii · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.ii only.

2 (a) A human sperm cell has a tail, a mid-piece containing organelles and a head with a nucleus.

(i) Give the function of the tail.

(1)

(ii) Which of these, found in the mid-piece, release energy?

(1)

- ☒ **A** chromosomes
- ☒ **B** endoplasmic reticulum
- ☒ **C** mitochondria
- ☒ **D** ribosomes

(iii) The process of meiosis is involved in the production of sperm.

Explain the importance of meiosis in sperm production.

(3)

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5. Question 2 · part b.ii · 2 marks

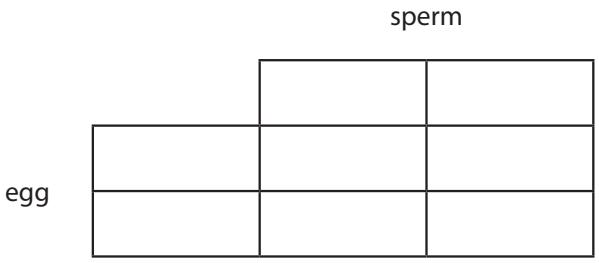
January 2023 Paper 1

Further practice · Attempt part b.ii only.

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(b) (i) Complete the genetic diagram to show how the sex of a person is determined by the chromosomes in the nucleus of the egg and the nucleus of the sperm.

(3)



(ii) The sperm head has to enter the egg cell so the nuclei can fuse.

Describe how the sperm head enters the egg cell.

(2)

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(Total for Question 2 = 10 marks)

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Working and answer

6. Question 3 · part b.iii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.iii only.

3 (a) Describe what is meant by a specialised cell.

(2)

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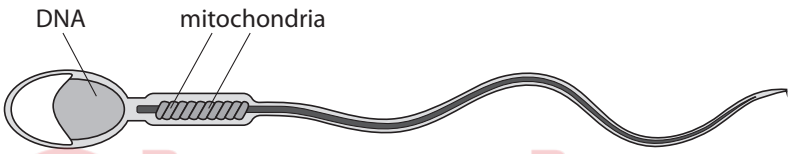
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(b) The sperm cell is an example of a specialised cell.

The diagram shows a sperm cell.

NOT TO SCALE



(i) The sperm cell in the diagram has an actual length of 0.0060 mm.

The length of the image of the sperm cell is 84 mm.

Calculate the magnification of the sperm cell.

(2)

magnification = × .....

(ii) The head of the sperm contains DNA.

Give the difference in the amount of DNA found in the head of the sperm and in a nucleus of a body cell.

(1)

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6. Question 3 · part b.iii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.iii only.

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(iii) Explain why the sperm cell has many mitochondria.

(2)

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(iv) A sperm cell travels at a speed of 0.070 mm/s.

It travels a distance of 180 mm to fertilise an egg.

Use this equation to calculate the time it takes for the sperm to reach the egg.

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Give your answer in minutes.

(3)

time taken = ... minutes

(c) An electron microscope can be used to view structures in specialised cells.

Explain why an electron microscope is more useful than a light microscope to view cell structure.

(2)

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(Total for Question 3 = 12 marks)

6. Question 3 · part b.iii · 2 marks · Working and answer

Working and answer

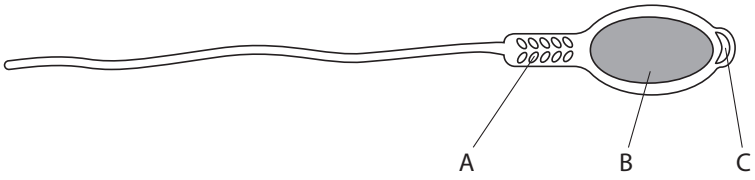
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7. Question 6 · part a.i · 3 marks

November 2023 Paper 2

Further practice · Attempt part a.i only.

6 The diagram shows a human sperm.



(a) (i) Name the parts A, B and C.

(3)

A .....

B .....

C .....

(ii) Describe the function of part C.

(3)

.....

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(iii) The actual length of the sperm is 50.0 micrometres (μm).

Calculate the magnification of the diagram.

[1 mm = 1000 μm]

(4)

magnification = ...

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P 7 4 4 9 4 A 0 1 4 2 0

7. Question 6 · part a.i · 3 marks · Working and answer

Working and answer

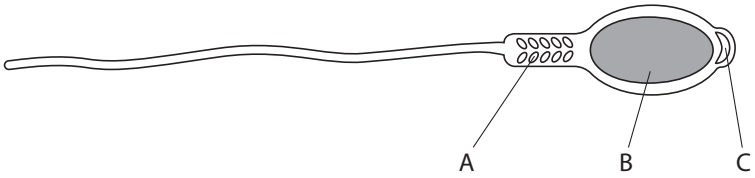
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8. Question 6 · part b · 3 marks

November 2023 Paper 2

Further practice · Attempt part b only.

6 The diagram shows a human sperm.



(a) (i) Name the parts A, B and C.

(3)

A .....

B .....

C .....

(ii) Describe the function of part C.

(3)

.....

.....

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.....

(iii) The actual length of the sperm is 50.0 micrometres (μm).

Calculate the magnification of the diagram.

[1 mm = 1000 μm]

(4)

magnification = ...

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P 7 4 4 9 4 A 0 1 4 2 0

8. Question 6 · part b · 3 marks

November 2023 Paper 2

Further practice · Attempt part b only.

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(b) Describe the function of the structure labelled A in the process of fertilisation.

(3)

(Total for Question 6 = 13 marks)

Working and answer

OLevels.com SCHOOL

Practice workbook · State the symptoms and effects of inherited conditions

Effects of polydactyly, haemophilia and cystic fibrosis on the body.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-STATE-INHERITED-CONDITION-EFFECTS

1. Question 6 · part b.ii · 4 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.ii only.

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6 (a) Individuals vary in their phenotypes.

State what is meant by the phenotype of an individual.

(1)

(b) (i) Cystic fibrosis is a disorder that affects epithelial membranes.

Describe what causes an individual to have cystic fibrosis.

(2)

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(ii) Describe the effects of cystic fibrosis on the health of an individual.

(4)

1. Question 6 · part b.ii · 4 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.ii only.

(iii) Explain why two parents who do not have cystic fibrosis can have a child who has cystic fibrosis.

(2)

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(Total for Question 6 = 9 marks)

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Working and answer

2. Question 3 · part a.iii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.iii only.

3 (a) Polydactyly is an inherited condition.

There are several faulty genes on different chromosomes that can cause polydactyly.

(i) State which part of a cell contains the most chromosomes.

(1)

(ii) Genes exist in alternative forms.

State the name given to an alternative form of a gene.

(1)

(iii) Give one symptom of polydactyly.

(1)

(b) Polydactyly is caused by the dominant form of the gene.

The table shows the genotypes of two parents, R and S.

Parent	Genotype
female (R)	Pp
male (S)	pp

(i) Name the term used to describe the genotype of parent R.

(1)

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2. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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3. Question 9 · part b.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part b.ii only.

9 (a) There are four main blood groups.

One of these groups is known as type O.

(i) Name the other three blood groups.

(2)

(ii) A person with blood type O needs a blood transfusion.

Explain which blood type this person should receive in a transfusion.

(5)

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3. Question 9 · part b.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part b.ii only.

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(b) Haemophilia is a sex-linked genetic condition that affects the blood.

(i) Describe what is meant by a sex-linked genetic condition.

(2)

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(ii) State one effect that haemophilia will have on an individual.

(1)

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(Total for Question 9 = 10 marks)

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Working and answer

4. Question 4 · part c.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c.ii only.

4 (a) A student does a survey of eye colour in a sample of people.

The table shows a tally of the number of people with each eye colour in the sample.

Blue	Green	Brown	Hazel	Grey
𝍸 𝍸 𝍸 𝍸 𝍸	𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸	III

(i) Complete the table by giving the missing information.

(1)

Number of people with each eye colour				
Blue	**Green**	**Brown**	**Hazel**	**Grey**
25				3

(ii) Calculate the percentage of people with brown eyes in the sample.

(2)

percentage = ... %

(b) Eye colour is determined by alleles inherited from parents.

(i) Which of these gives the definition of an allele?

(1)

- ☒ **A** two dominant genes in a genotype
- ☒ **B** two recessive genes in a genotype
- ☒ **C** different genes in a genotype
- ☒ **D** alternative forms of the same gene in a genotype

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P 7 4 4 9 5 A 0 1 2 2 8

4. Question 4 · part c.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c.ii only.

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(ii) Explain why diploid cells contain two alleles for each characteristic.

(4)

(c) Cystic fibrosis is a condition that is also determined by alleles that are inherited from parents.

(i) Give the possible genotypes of two parents that have a child with cystic fibrosis.

(2)

(ii) Give one symptom of cystic fibrosis.

(1)

(Total for Question 4 = 11 marks)

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4. Question 4 · part c.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain oral rehydration and treatment of diarrhoeal disease

Oral rehydration therapy for cholera or Ebola, plus antibiotics where the disease is bacterial.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: HBI-ORAL-REHYDRATION-TREATMENT

1. Question 7 · part c · 4 marks

June 2022 Paper 1R

Worked-example retry · Attempt part c only.

7 Cholera is a disease that causes chronic water loss from the body.

(a) Name the type of microorganism that causes cholera.

(1)

(b) Describe how cholera is transmitted from one person to another.

(2)

(c) Explain how a person with cholera can be treated for the disease.

(4)

(Total for Question 7 = 7 marks)

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1. Question 7 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part d · 3 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

3 Ebola is a disease that has become an epidemic in some countries.

(a) Which of these causes Ebola?

(1)

- ☒ **A** a bacterium
- ☒ **B** a mosquito
- ☒ **C** a protozoan
- ☒ **D** a virus

(b) State three precautions that can be taken to prevent the spread of Ebola.

(3)

1 .....

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2 .....

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3 .....

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(c) Symptoms of Ebola only appear two to three weeks after infection.

Explain why the symptoms do not appear immediately.

(3)

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P 6 0 1 8 6 A 0 6 2 8

2. Question 3 · part d · 3 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

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(d) Oral rehydration is a method used to treat the symptoms of Ebola.

Explain why this method is an effective treatment.

(3)

(e) Fruit bats are thought to be a natural reservoir for the Ebola pathogen.

Suggest what is meant by the term **reservoir**.

(2)

(Total for Question 3 = 12 marks)

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2. Question 3 · part d · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 8 · part b · 3 marks

June 2024 Paper 1R

Further practice · Attempt part b only.

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8 (a) Cholera is a disease that kills approximately 95 000 people each year worldwide.

(i) Name the type of pathogen that causes cholera.

(1)

.....

(ii) State how cholera is transmitted from one person to another.

(1)

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.....

(b) Describe a method used to treat a person with cholera.

(3)

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(Total for Question 8 = 5 marks)

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OLevels.com SCHOOL

Working and answer

Practice workbook · Explain sewage treatment, pit latrine design and decomposer roles

Stages of a sewage works, aerobic and anaerobic microbes, biogas and sludge, decomposers recycling nutrients, and how pit latrines are sited, built and covered.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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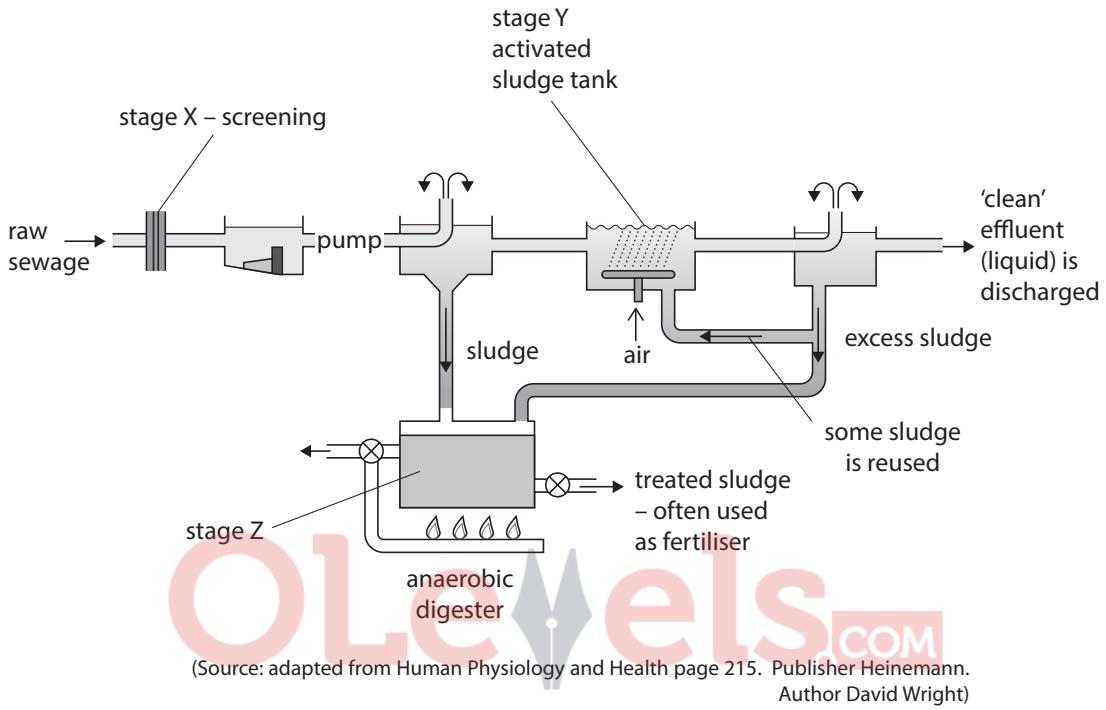
Type ID: HBI-SANITATION-SEWAGE-AND-LATRINES

1. Question 8 · part a.ii · 2 marks

June 2024 Paper 1

Worked-example retry · Attempt part a.ii only.

8 (a) The diagram represents a modern sewage works.



(Source: adapted from Human Physiology and Health page 215. Publisher Heinemann. Author David Wright)

(i) Describe the process that happens at stage X.

(2)

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1. Question 8 · part a.ii · 2 marks

June 2024 Paper 1

Worked-example retry · Attempt part a.ii only.

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(ii) Explain why air is added at stage Y.

(2)

(iii) Biogas is produced at stage Z.

Name the main gas found in biogas.

(1)

(b) Explain why treated sludge from a sewage works can be used as a fertiliser.

(2)

(Total for Question 8 = 7 marks)

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1. Question 8 · part a.ii · 2 marks · Working and answer

Working and answer

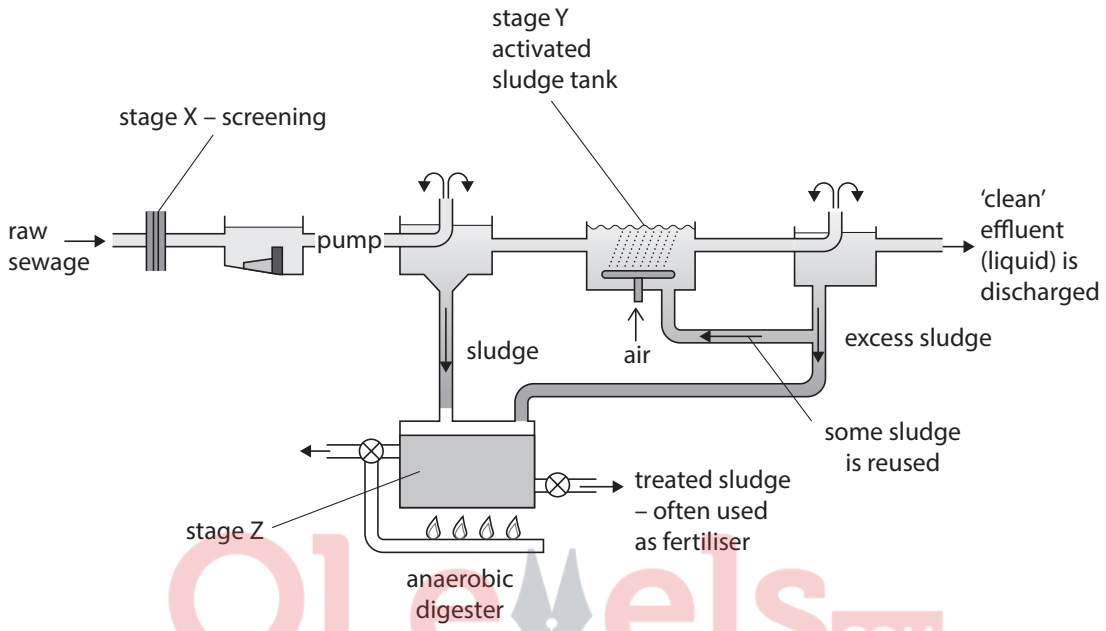
OLevels.com SCHOOL

2. Question 8 · part a.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part a.i only.

8 (a) The diagram represents a modern sewage works.



(Source: adapted from Human Physiology and Health page 215. Publisher Heinemann. Author David Wright)

(i) Describe the process that happens at stage X.

(2)

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P 7 4 6 2 6 R A 0 2 0 2 4

2. Question 8 · part a.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part a.i only.

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(ii) Explain why air is added at stage Y.

(2)

(iii) Biogas is produced at stage Z.

Name the main gas found in biogas.

(1)

(b) Explain why treated sludge from a sewage works can be used as a fertiliser.

(2)

(Total for Question 8 = 7 marks)

OLevels.com SCHOOL

2. Question 8 · part a.i · 2 marks · Working and answer

Working and answer

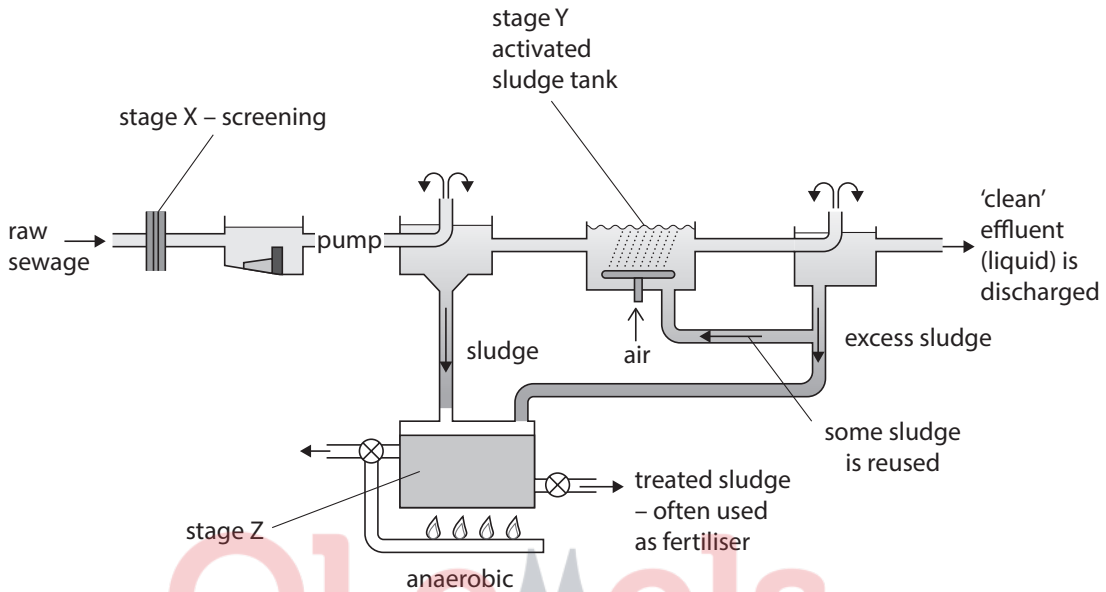
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3. Question 8 · part a.iii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.iii only.

8 (a) The diagram represents a modern sewage works.



(Source: adapted from Human Physiology and Health page 215. Publisher Heinemann. Author David Wright)

(i) Describe the process that happens at stage X.

(2)

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3. Question 8 · part a.iii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.iii only.

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(ii) Explain why air is added at stage Y.

(2)

(iii) Biogas is produced at stage Z.

Name the main gas found in biogas.

(1)

(b) Explain why treated sludge from a sewage works can be used as a fertiliser.

(2)

(Total for Question 8 = 7 marks)

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3. Question 8 · part a.iii · 1 mark · Working and answer

Working and answer

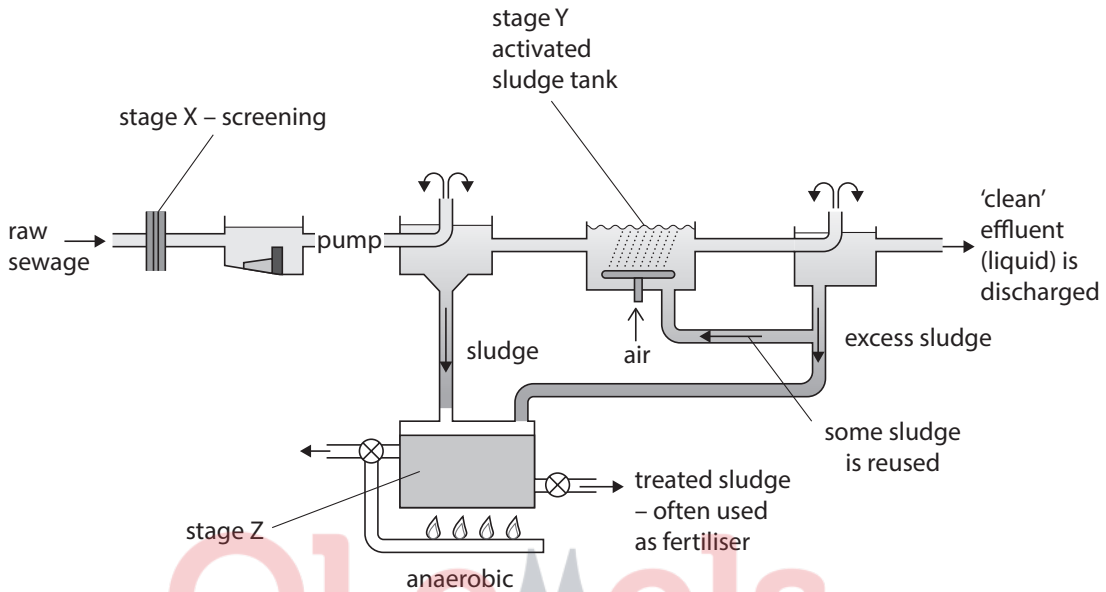
OLevels.com SCHOOL

4. Question 8 · part b · 2 marks

June 2024 Paper 1

Further practice · Attempt part b only.

8 (a) The diagram represents a modern sewage works.



(Source: adapted from Human Physiology and Health page 215. Publisher Heinemann. Author David Wright)

(i) Describe the process that happens at stage X.

(2)

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P 7 4 6 2 6 R A 0 2 0 2 4

4. Question 8 · part b · 2 marks

June 2024 Paper 1

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

(ii) Explain why air is added at stage Y.

(2)

(iii) Biogas is produced at stage Z.

Name the main gas found in biogas.

(1)

DO NOT WRITE IN THIS AREA

(b) Explain why treated sludge from a sewage works can be used as a fertiliser.

(2)

(Total for Question 8 = 7 marks)

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4. Question 8 · part b · 2 marks · Working and answer

Working and answer

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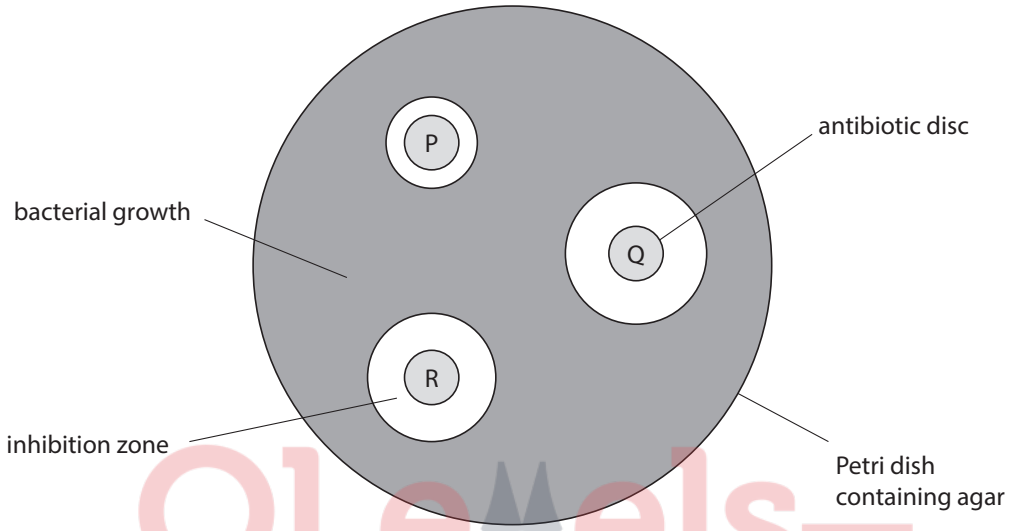
5. Question 5 · part b · 2 marks

June 2024 Paper 2

Further practice · Attempt part b only.

5 (a) A student investigates the action of three different antibiotics, P, Q and R, on the growth of bacteria.

The diagram shows the results of the investigation.



(i) Which is the independent variable in this investigation?

(1)

- ☒ **A** the mass of bacteria added
- ☒ **B** the size of the inhibition zone
- ☒ **C** the type of antibiotic added
- ☒ **D** the mass of agar in the Petri dish

(ii) Give two procedures that the student should use to reduce the risk of contamination by unwanted bacteria.

(2)

1 .....

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2 .....

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.....

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P 7 4 6 3 0 A 0 1 4 2 4

5. Question 5 · part b · 2 marks

June 2024 Paper 2

Further practice · Attempt part b only.

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(iii) Explain which is the most effective antibiotic.

(2)

(iv) The student concludes the bacteria are resistant to antibiotic P.

Explain how antibiotic resistance occurs in bacteria.

(4)

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(b) Decomposers are microorganisms such as bacteria and fungi that do not cause disease.

Describe the role of these microorganisms in the environment.

(2)

(Total for Question 5 = 11 marks)

5. Question 5 · part b · 2 marks · Working and answer

Working and answer

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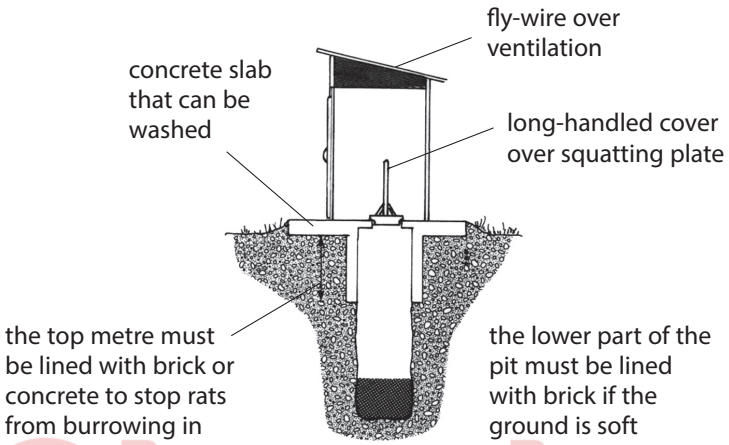
6. Question 5 · part a.i · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.i only.

5 Pit latrines can be used to get rid of urine and faeces. They can be used where the soil is permeable to water and are always placed downhill from wells.

The diagram shows a pit latrine.



(a) (i) Explain why a pit latrine should be placed downhill from a well. **(2)**

(ii) Give the reason why pit latrines must be placed in soil that is permeable to water. **(1)**

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P 7 2 5 8 5 A 0 1 2 2 0

6. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

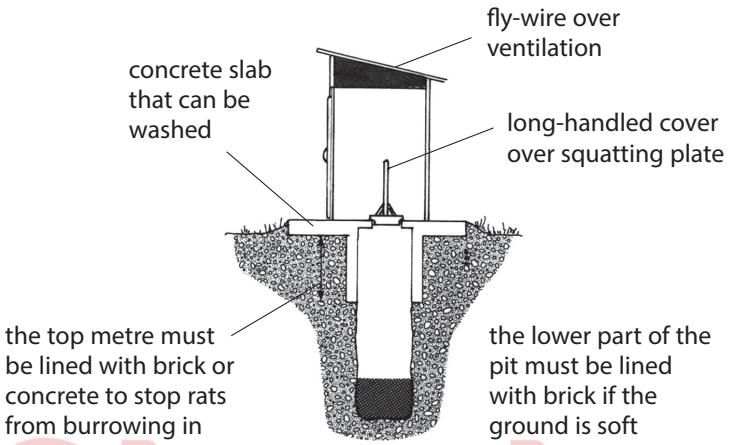
7. Question 5 · part a.ii · 1 mark

June 2023 Paper 1

Further practice · Attempt part a.ii only.

5 Pit latrines can be used to get rid of urine and faeces. They can be used where the soil is permeable to water and are always placed downhill from wells.

The diagram shows a pit latrine.



(a) (i) Explain why a pit latrine should be placed downhill from a well. **(2)**

(ii) Give the reason why pit latrines must be placed in soil that is permeable to water. **(1)**

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7. Question 5 · part a.ii · 1 mark

June 2023 Paper 1

Further practice · Attempt part a.ii only.

(iii) Explain why, when a pit latrine is not in use, the latrine should have a cover and why the cover should have a long handle.

(5)

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7. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

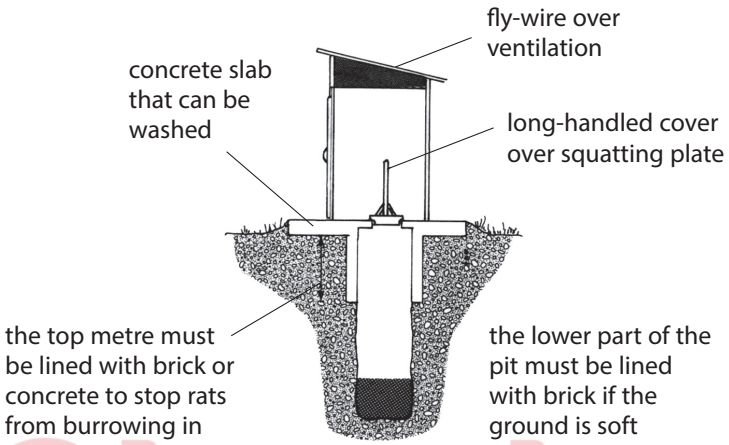
8. Question 5 · part a.iii · 5 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

5 Pit latrines can be used to get rid of urine and faeces. They can be used where the soil is permeable to water and are always placed downhill from wells.

The diagram shows a pit latrine.



(a) (i) Explain why a pit latrine should be placed downhill from a well. **(2)**

(ii) Give the reason why pit latrines must be placed in soil that is permeable to water. **(1)**

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8. Question 5 · part a.iii · 5 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

(iii) Explain why, when a pit latrine is not in use, the latrine should have a cover and why the cover should have a long handle.

(5)

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8. Question 5 · part a.iii · 5 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

(b) The concrete slab shown in the diagram is washed regularly with an antibacterial solution.

A scientist wants to investigate two antibacterial solutions to see which solution is more effective.

Design an investigation the scientist could use to find out which of the two antibacterial solutions is more effective.

Include experimental details in your answer and write in full sentences.

(7)

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(Total for Question 5 = 15 marks)

8. Question 5 · part a.iii · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Describe virus structure and reproduction

The protein coat and nucleic acid of viruses and how they reproduce inside host cells.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: HBI-VIRUS-STRUCTURE-REPRODUCTION

1. Question 9 · part b · 6 marks

June 2022 Paper 2

Worked-example retry · Attempt part b only.

9 (a) Hepatitis is a disease of the liver.

This disease can be caused by drinking too much alcohol.

Explain the damaging effects of alcohol on the liver.

(3)

(b) Hepatitis B is one type of virus that can cause hepatitis.

Describe the reproduction of a hepatitis B virus.

(6)

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OLevels.COM SCHOOL

1. Question 9 · part b · 6 marks

June 2022 Paper 2

Worked-example retry · Attempt part b only.

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(c) Vaccinations can be used to prevent the spread of hepatitis.

Explain how vaccines can help people become immune to hepatitis.

(3)

(Total for Question 9 = 12 marks)

OLevels.com SCHOOL

1. Question 9 · part b · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 2 · part a.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.i only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Treatment of Ebola

Ebola is caused by an RNA virus. There is no proven treatment available for Ebola. However, whole blood collected from someone who has recovered from infection has been used as a treatment. The results have been promising in a small group of Ebola patients.

Guidance for national health authorities and blood transfusion services describes the stages needed to collect whole blood (CWB) or plasma (CP) from Ebola-recovered patients. This can be used for transfusion to patients as a treatment for early Ebola.

The guidance states how to

- identify people who have recovered from Ebola as potential blood donors
- screen donors' blood for A, B, O and Rhesus groupings
- screen donors' blood for transfusion-transmissible infections
- collect blood and care for donors
- obtain agreement of Ebola patients for the treatment
- identify patient's blood grouping
- store and transport CWB and CP to the places where transfusions are to be given
- select Ebola patients for transfusion
- transfuse the blood into the patient

Donated CWB should never be frozen and should be stored between 2 °C and 6 °C.

CWB and CP donations should be transfused to the Ebola patients using standard transfusion procedures.

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2. Question 2 · part a.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.i only.

(a) (i) Explain what is meant by an RNA virus (line 1).

(2)

(ii) The table lists five diseases. Put a tick (✓) next to the diseases that should be tested for in the donors' blood.

(2)

Disease	Blood tested (✓)
anaemia	
cystic fibrosis	
gonorrhoea	
HIV	
scurvy	

(iii) Explain why it is necessary to identify the blood group of the blood collected from donors who have recovered from Ebola (line 9).

(3)

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2. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 8 · part a.i · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a.i only.

8 (a) (i) Describe the structure of a virus.

(2)

(ii) Viruses attach to cells of various body tissues.

The virus uses the host cell's genetic material to produce new viral proteins.

Describe how the host cell carries out protein synthesis to produce new viral proteins.

(5)

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3. Question 8 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain genetic and environmental causes of variation

Whether a feature varies because of inherited alleles, surroundings or both.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: HBI-EXPLAIN-GENETIC-AND-ENVIRONMENTAL-VARIATION

1. Question 3 · part d · 3 marks

November 2021 Paper 2

Worked-example retry · Attempt part d only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Sweating is normal when we are hot. It is part of our natural cooling mechanism. If you sweat visibly when you do not need to cool down, it can be a problem.

The sweating process is controlled by the brain, which sends signals down the nerves to the sweat glands.

Sweating is crucial to survival. We sweat to control body temperature and the sweat glands in our armpits can produce several litres of sweat in 24 hours. How much we sweat varies hugely from individual to individual and even from day to day.

Hyperhidrosis occurs when some areas of the body start sweating excessively. This affects about 1% of the population. There are two main types. The most common type is primary focal hyperhidrosis, which affects many parts of the body. There is no known cause, but it usually begins in childhood and often runs in families. Another type is secondary hyperhidrosis, which often doesn't begin until after the teenage years, and usually has an underlying medical cause. These causes can include diabetes, menopause or chronic infection. It can also be a result of eating certain foods. Stress can play a major role too.

(a) Which part of the brain controls sweating (line 4)?

(1)

- ☒ **A** cerebellum
- ☒ **B** cerebral hemispheres
- ☒ **C** hypothalamus
- ☒ **D** pituitary gland

(b) Which type of nerve sends signals to the sweat glands (lines 4 and 5)?

(1)

- ☒ **A** connector
- ☒ **B** motor
- ☒ **C** relay
- ☒ **D** sensory

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1. Question 3 · part d · 3 marks

November 2021 Paper 2

Worked-example retry · Attempt part d only.

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(c) Describe how sweating controls body temperature (lines 6 and 7).

(3)

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(d) Explain the evidence in the passage that indicates primary focal hyperhidrosis is a genetic condition.

(3)

OLevels.com SCHOOL

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(e) Secondary hyperhidrosis can be caused by diabetes.

Give one risk factor for diabetes (line 16).

(1)

1. Question 3 · part d · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part b · 3 marks

November 2020 Paper 1

Further practice · Attempt part b only.

3 There are many factors that affect the height of a human.

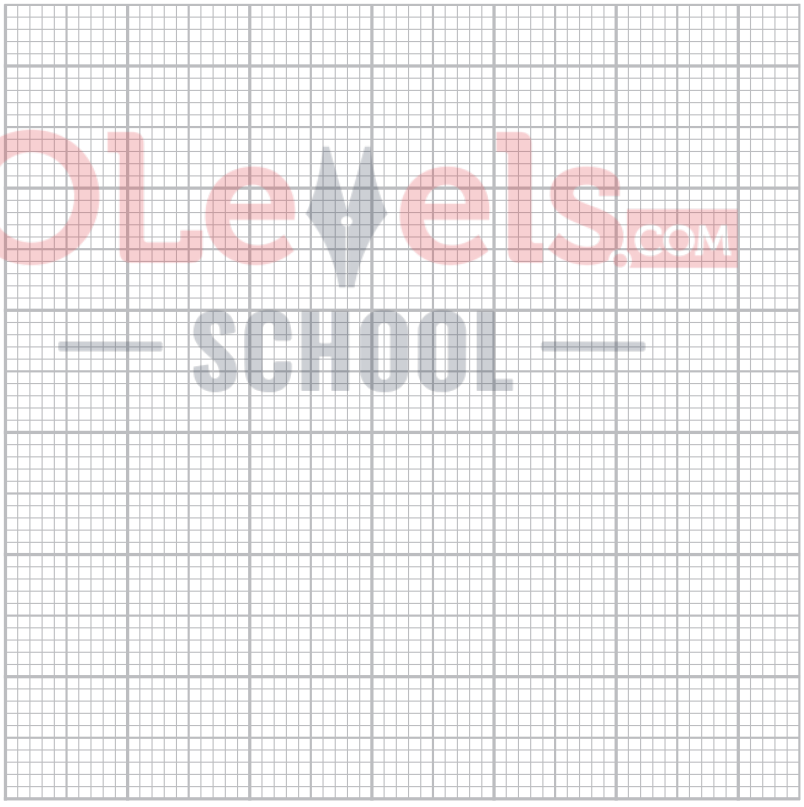
(a) The heights of a group of students are measured.

The table shows the results.

Height in cm	150–151	152–153	154–155	156–157	158–159	> 160
Number of students	4	14	25	20	10	4

(i) Plot a bar chart of the results.

(5)



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2. Question 3 · part b · 3 marks

November 2020 Paper 1

Further practice · Attempt part b only.

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(ii) Describe the trend shown in this bar chart.

(2)

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(b) State three factors that could affect the heights of the students.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(Total for Question 3 = 10 marks)

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OLevels.com SCHOOL

2. Question 3 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 5 · part d · 2 marks

June 2022 Paper 2R

Further practice · Attempt part d only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Milk contains a carbohydrate called lactose. Lactose is not absorbed by the epithelium of the ileum or duodenum.

Lactose is a disaccharide which is broken down into glucose and galactose by the enzyme lactase. Babies produce large amounts of lactase but in some people the gene for producing lactase is switched off as they get older. When this gene is switched off people then become lactose-intolerant which means they cannot digest any lactose that is taken into the gut.

The number of people with lactose intolerance differs across the world. People in some countries are often tolerant because their ancestors have evolved tolerance because of their diet.

In a lactose intolerant person, any lactose taken into the gut will collect and ferment, forming methane and carbon dioxide plus other soluble products. These other soluble products lower the water potential in the gut.

Symptoms of lactose intolerance include diarrhoea, calcium deficiency and vitamin D deficiency.

(a) Suggest what is meant by the term disaccharide. (lines 3 to 4) **(2)**

(b) Give a reason why lactose is not absorbed by the epithelium. (lines 1 to 2) **(1)**

(c) (i) Explain what is meant by a gene being switched off. (lines 5 to 6) **(3)**

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3. Question 5 · part d · 2 marks

June 2022 Paper 2R

Further practice · Attempt part d only.

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(ii) Describe the effects on a person when the gene for producing lactase is switched off. (lines 3 to 7)

(2)

(d) Suggest how people in some parts of the world have evolved tolerance because of their diet. (lines 8 to 10)

(2)

OLevels.COM SCHOOL

(e) Explain how the symptoms of lactose intolerance are caused. (lines 11 to 15)

(4)

(Total for Question 5 = 14 marks)

3. Question 5 · part d · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Interpret human growth data, proportions and percentile charts

Height tables, body proportion diagrams and percentile growth charts from birth to adulthood.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: HBI-INTERPRET-HUMAN-GROWTH-DATA

1. Question 8 · part c.ii · 3 marks

January 2023 Paper 2

Worked-example retry · Attempt part c.ii only.

8 (a) When women cannot conceive naturally, they can have fertility treatment.

Women having fertility treatment can be given the hormones FSH and LH.

(i) Give a role of LH in women.

(1)

(ii) Giving birth to twins or triplets is described as a multiple birth.

Explain how the use of FSH to treat infertility could increase the chances of a multiple birth.

(3)

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OLevels.com SCHOOL

24

1. Question 8 · part c.ii · 3 marks

January 2023 Paper 2

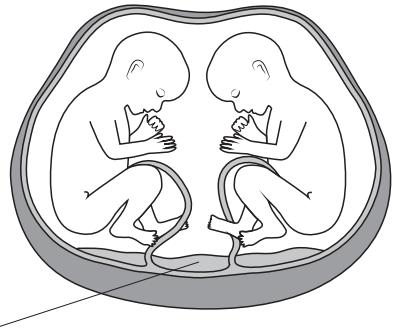
Worked-example retry · Attempt part c.ii only.

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(b) Some identical twins share a placenta, as shown in the diagram.



(i) Twins that share a placenta are often born with a lower mass than twins that have separate placentas.

Explain why twins sharing a placenta have a lower mass when they are born.

(3)

OLevels.com SCHOOL

(ii) Identical twins form when the embryo splits early in pregnancy.

Explain why the twins are genetically identical.

(2)

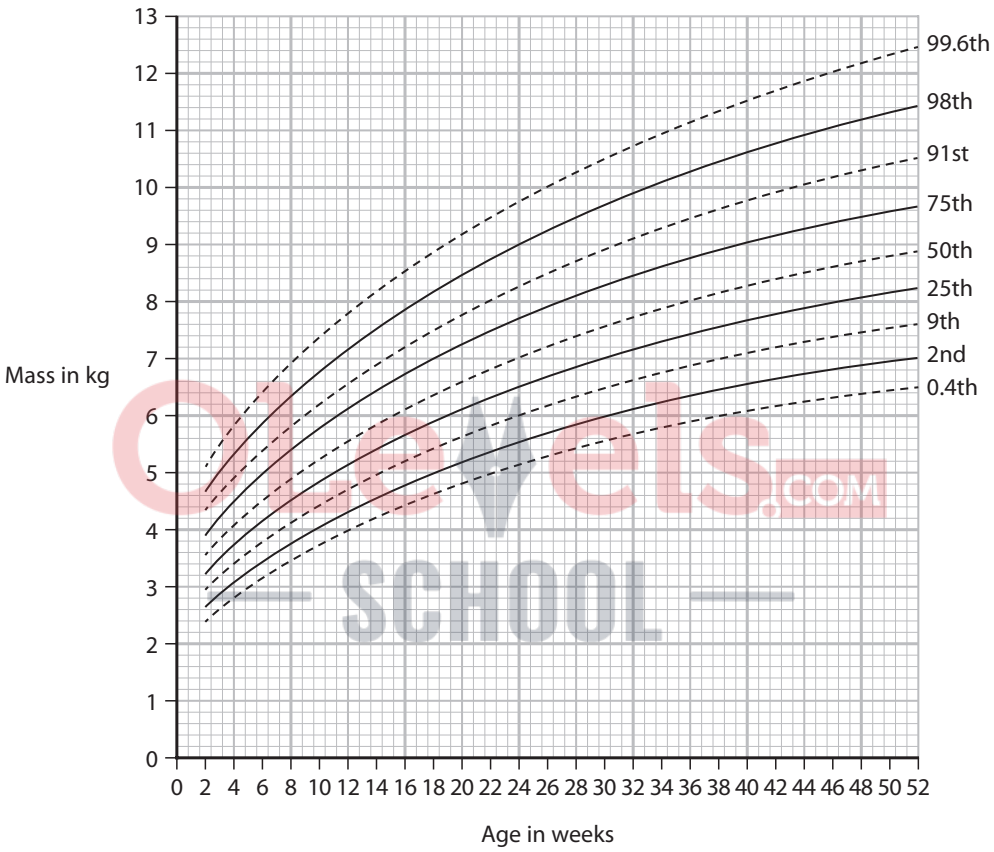
1. Question 8 · part c.ii · 3 marks

January 2023 Paper 2

Worked-example retry · Attempt part c.ii only.

(c) When babies are young, their growth is monitored using a percentile chart.

The diagram shows a percentile chart for baby girls.



(i) Determine the growth percentile of a six-week-old baby girl with a mass of 5 kg.

(1)

percentile = ...

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1. Question 8 · part c.ii · 3 marks

January 2023 Paper 2

Worked-example retry · Attempt part c.ii only.

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(ii) Describe how percentile charts are used to identify concerns about the growth of a baby.

(3)

(Total for Question 8 = 13 marks)

OLevels.com SCHOOL

Working and answer

2. Question 5 · part a.ii · 2 marks

June 2019 Paper 1

Further practice · Attempt part a.ii only.

5 A teacher wants to calculate the body mass index (BMI) of a number of students.

(a) The teacher measures the height of students from four age groups.

There are two male and two female students in each age group.

The table shows the data collected.

Age in years	Height of student in cm				Mean height of students in cm
	Male		Female		
12	146.2	148.9	142.8	144.2	145.5
14	165.7	166.4	161.9	164.0	164.5
16	175.9	178.5	166.3	167.6	
18	180.9	181.3	171.2	174.3	176.9

(i) Calculate the mean height of the students aged 16.

Give your answer to one decimal place.

(2)

OLevels.com SCHOOL

mean height = ... cm

(ii) State two conclusions that the teacher could make from the data in the table.

(2)

1 .....

.....

2 .....

.....

(iii) Give two ways the teacher could improve his investigation.

(2)

1 .....

.....

2 .....

.....

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2. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

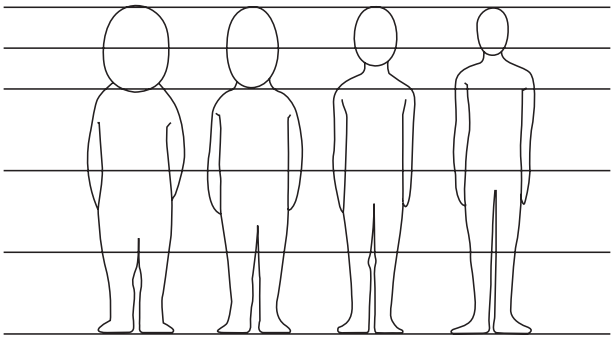
OLevels.com SCHOOL

3. Question 4 · part b · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

4 The diagram shows the changes in the proportions of different parts of the body of a person over a period of 18 years.



newborn 2 years 7 years 18 years

(a) (i) Estimate the proportion of the body that is the head in a newborn baby.

(1)

proportion = ...

(ii) Estimate the proportion of the body that is the legs in an 18-year old person.

(1)

proportion = ...

(b) Describe two main changes in the proportions of the body between birth and 18 years.

(2)

1 .....

.....

2 .....

.....

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3. Question 4 · part b · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

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(c) Suggest why it is useful to measure growth of a body part as a proportion of the whole body length.

(2)

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(Total for Question 4 = 6 marks)

Working and answer

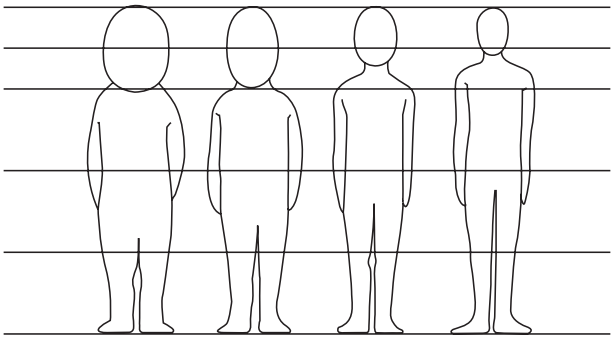
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4. Question 4 · part c · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c only.

4 The diagram shows the changes in the proportions of different parts of the body of a person over a period of 18 years.



newborn 2 years 7 years 18 years

(a) (i) Estimate the proportion of the body that is the head in a newborn baby.

(1)

proportion = ...

(ii) Estimate the proportion of the body that is the legs in an 18-year old person.

(1)

proportion = ...

(b) Describe two main changes in the proportions of the body between birth and 18 years.

(2)

1 .....

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2 .....

.....

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4. Question 4 · part c · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c only.

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(c) Suggest why it is useful to measure growth of a body part as a proportion of the whole body length.

(2)

.....

.....

.....

.....

(Total for Question 4 = 6 marks)

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain how the placenta, cord and hormones support a fetus

Placental exchange, umbilical cord function and progesterone maintaining the lining during pregnancy.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: HBI-EXPLAIN-PLACENTA-AND-PREGNANCY-SUPPORT

1. Question 8 · part b.i · 3 marks

January 2023 Paper 2

Worked-example retry · Attempt part b.i only.

8 (a) When women cannot conceive naturally, they can have fertility treatment.

Women having fertility treatment can be given the hormones FSH and LH.

(i) Give a role of LH in women.

(1)

(ii) Giving birth to twins or triplets is described as a multiple birth.

Explain how the use of FSH to treat infertility could increase the chances of a multiple birth.

(3)

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OLevels.COM SCHOOL

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1. Question 8 · part b.i · 3 marks

January 2023 Paper 2

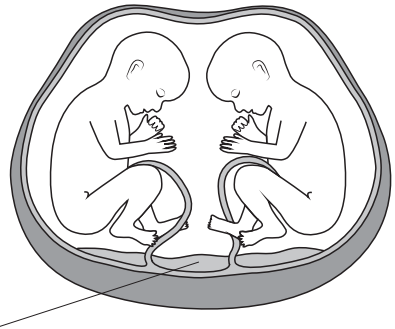
Worked-example retry · Attempt part b.i only.

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(b) Some identical twins share a placenta, as shown in the diagram.



(i) Twins that share a placenta are often born with a lower mass than twins that have separate placentas.

Explain why twins sharing a placenta have a lower mass when they are born.

(3)

OLevels.com SCHOOL

(ii) Identical twins form when the embryo splits early in pregnancy.

Explain why the twins are genetically identical.

(2)

1. Question 8 · part b.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 6 · part d · 2 marks

June 2023 Paper 2R

Further practice · Attempt part d only.

6 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Pregnancy begins with the formation of a zygote. This stage is known as conception. The zygote develops into an embryo. By day five the embryo is transformed into a blastocyst, a hollow ball of about 100 genetically identical cells. These cells have not yet differentiated and they can provide a source of embryonic stem cells for use in medical research. Embryos produced following IVF treatment can be used as a source of embryonic stem cells.

In the second week following conception, the blastocyst begins the process of implantation and releases a hormone, called hCG. This hormone maintains the corpus luteum, a structure formed in the ovary following ovulation. After implantation the amniotic sac forms, which is where the blastocyst and later the fetus develop.

In the fourth and fifth week after conception, the umbilical cord forms. This connects the embryo to the placenta by week eight. The placenta is not fully developed until week ten when it begins to produce progesterone and oestrogen. All organ systems are present at this time in pregnancy.

Fetal development begins at about month four and it is possible to distinguish males from females at this time. In males the testes, which develop in the abdominal cavity, release testosterone. Female characteristics are produced in the fetus if testosterone is not released at the stage.

Hormones are important in starting labour and the birth process, about nine months after conception. The cervix dilates, with the head of the baby pushing against the cervix to help the process of dilation. Once the child is born and breathing normally, the umbilical cord is tied and cut, separating the child from the placenta.

(a) Describe the process that occurs at conception. (lines 1 and 2)

(2)

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2. Question 6 · part d · 2 marks

June 2023 Paper 2R

Further practice · Attempt part d only.

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(b) Describe the process that forms a blastocyst. (lines 2 to 4)

(4)

OLevels.com SCHOOL

(c) Which is a reason why cells taken from a blastocyst can be useful in scientific research?

(1)

- ☒ **A** the cells are already differentiated
- ☒ **B** the cells are able to differentiate into both sex cells and body cells
- ☒ **C** the cells are able to differentiate into any type of sex cell
- ☒ **D** the cells are able to differentiate into any type of body cell

2. Question 6 · part d · 2 marks

June 2023 Paper 2R

Further practice · Attempt part d only.

(d) Explain why it is important that hCG maintains the corpus luteum. (lines 7 to 9)

(2)

(e) Explain the importance of the development that takes place in the fourth and fifth week after conception. (lines 12 and 13)

(2)

(f) Suggest the role of testosterone in the development of the fetus. (lines 17 to 19)

(1)

(g) Which change occurs during the birth process? (line 20)

(1)

- ☒ **A** oestrogen levels decrease
- ☒ **B** progesterone levels increase
- ☒ **C** release of oxytocin from the pituitary gland
- ☒ **D** release of hCG from the placenta

(Total for Question 6 = 13 marks)

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OLevels.com SCHOOL

2. Question 6 · part d · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 6 · part e · 2 marks

June 2023 Paper 2R

Further practice · Attempt part e only.

6 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Pregnancy begins with the formation of a zygote. This stage is known as conception. The zygote develops into an embryo. By day five the embryo is transformed into a blastocyst, a hollow ball of about 100 genetically identical cells. These cells have not yet differentiated and they can provide a source of embryonic stem cells for use in medical research. Embryos produced following IVF treatment can be used as a source of embryonic stem cells.

In the second week following conception, the blastocyst begins the process of implantation and releases a hormone, called hCG. This hormone maintains the corpus luteum, a structure formed in the ovary following ovulation. After implantation the amniotic sac forms, which is where the blastocyst and later the fetus develop.

In the fourth and fifth week after conception, the umbilical cord forms. This connects the embryo to the placenta by week eight. The placenta is not fully developed until week ten when it begins to produce progesterone and oestrogen. All organ systems are present at this time in pregnancy.

Fetal development begins at about month four and it is possible to distinguish males from females at this time. In males the testes, which develop in the abdominal cavity, release testosterone. Female characteristics are produced in the fetus if testosterone is not released at the stage.

Hormones are important in starting labour and the birth process, about nine months after conception. The cervix dilates, with the head of the baby pushing against the cervix to help the process of dilation. Once the child is born and breathing normally, the umbilical cord is tied and cut, separating the child from the placenta.

(a) Describe the process that occurs at conception. (lines 1 and 2)

(2)

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3. Question 6 · part e · 2 marks

June 2023 Paper 2R

Further practice · Attempt part e only.

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(b) Describe the process that forms a blastocyst. (lines 2 to 4)

(4)

OLevels.com SCHOOL

(c) Which is a reason why cells taken from a blastocyst can be useful in scientific research?

(1)

- ☒ **A** the cells are already differentiated
- ☒ **B** the cells are able to differentiate into both sex cells and body cells
- ☒ **C** the cells are able to differentiate into any type of sex cell
- ☒ **D** the cells are able to differentiate into any type of body cell

3. Question 6 · part e · 2 marks

June 2023 Paper 2R

Further practice · Attempt part e only.

(d) Explain why it is important that hCG maintains the corpus luteum. (lines 7 to 9)

(2)

(e) Explain the importance of the development that takes place in the fourth and fifth week after conception. (lines 12 and 13)

(2)

(f) Suggest the role of testosterone in the development of the fetus. (lines 17 to 19)

(1)

(g) Which change occurs during the birth process? (line 20)

(1)

- ☒ **A** oestrogen levels decrease
- ☒ **B** progesterone levels increase
- ☒ **C** release of oxytocin from the pituitary gland
- ☒ **D** release of hCG from the placenta

(Total for Question 6 = 13 marks)

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3. Question 6 · part e · 2 marks · Working and answer

Working and answer

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4. Question 4 · part b.ii · 3 marks

November 2023 Paper 2

Further practice · Attempt part b.ii only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Atorvastatin belongs to a group of medicines known as statins. Statins are medicines that regulate lipid levels in the blood. Statins are used when changes of diet and lifestyle have failed to lower blood cholesterol levels.

Statins can also be used to reduce the risk of heart disease when cholesterol levels are normal. People with liver disease or kidney problems should avoid taking statins. Women who are pregnant or are breastfeeding should also avoid taking statins.

In the United Kingdom approximately 7.5 million people take statins. This is 12% of the total population. However, up to 20% of these people stop taking statins because of adverse side effects such as muscle aches, fatigue, feeling sick and joint pains.

A recent study looked at patients who had stopped taking statins because of side effect symptoms. Patients were asked to take statins for four months, then placebo tablets (tablets that look like statins but contain no active substances) for four months followed by no tablets for four months. Every day the patients were asked how bad their symptoms were using a score from 0 to 100. The table shows the results.

Four-month treatment	Mean score out of 100
statins	16.3
placebo	15.4
no tablets	8.0

(a) Describe the changes in diet and lifestyle that could lower cholesterol. (Lines 2 and 3)

(2)

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P 7 4 4 9 4 A 0 8 2 0

4. Question 4 · part b.ii · 3 marks

November 2023 Paper 2

Further practice · Attempt part b.ii only.

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(b) (i) Explain why people with liver disease or kidney problems should avoid taking statins. (Lines 5 and 6)

(4)

(ii) Explain why women who are pregnant or breastfeeding should avoid taking statins. (Lines 6 and 7)

(3)

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(c) (i) Using the results of the study, calculate how many times greater the score for symptoms when taking statins is compared with the score when not taking tablets.

(1)

times greater = ...

4. Question 4 · part b.ii · 3 marks · Working and answer

Working and answer

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Practice workbook · Interpret antibacterial culture investigations

Petri dish and growth-graph investigations of antibiotics or bacterial growth: comparing inhibition zones or counts, variables and safe incubation.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ANTIBACTERIAL-CULTURE-INVESTIGATION

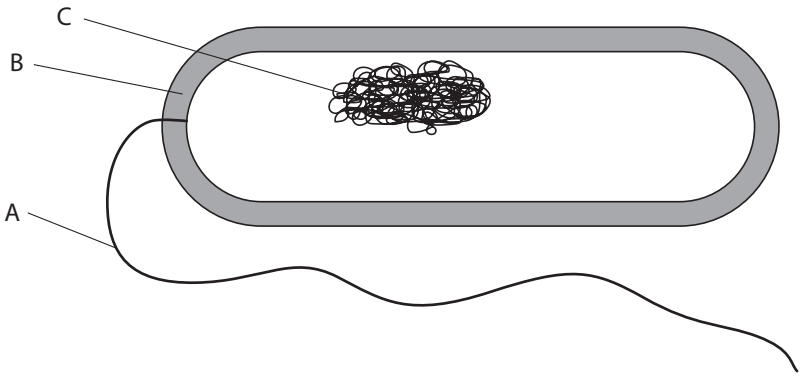
1. Question 8 · part b.ii · 3 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.ii only.

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8 The diagram shows a bacterial cell.



(a) (i) Name the parts labelled A, B and C.

(3)

A .....

B .....

C .....

(ii) Some types of bacteria cause disease.

Give the name of organisms that cause disease.

(1)

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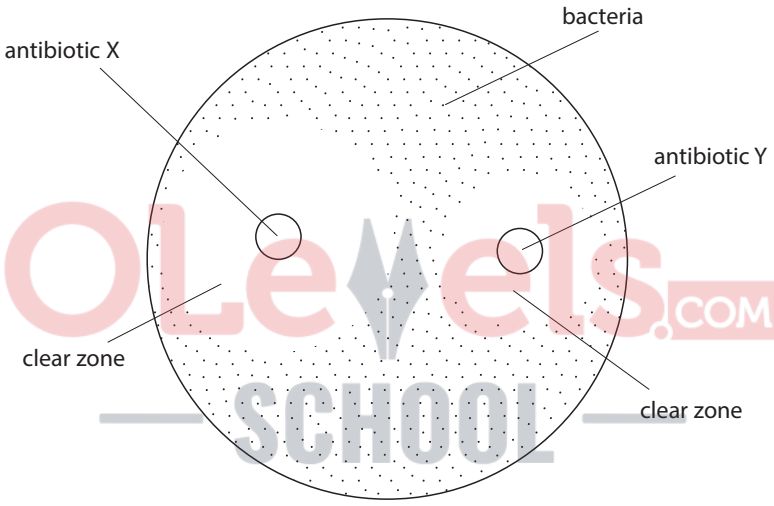
1. Question 8 · part b.ii · 3 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.ii only.

(b) A student uses this method to investigate whether antibiotic X is more effective than antibiotic Y.

- grow bacteria on nutrient agar in a Petri dish
- soak one disc of filter paper in antibiotic X and another disc in antibiotic Y
- place each disc on the agar in the Petri dish
- seal the Petri dish and leave for three days



(i) Describe what is meant by the term **antibiotic**.

(2)

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1. Question 8 · part b.ii · 3 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.ii only.

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(ii) Explain which antibiotic, X or Y, is more effective against the bacteria in the Petri dish.

(3)

(iii) Explain a suitable control that could be used in this investigation.

(3)

(Total for Question 8 = 12 marks)

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1. Question 8 · part b.ii · 3 marks · Working and answer

Working and answer

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2. Question 7 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

7 Various diseases are caused by microorganisms. Some diseases can be treated by using antibiotics.

(a) (i) Which of these is the name given to any microorganisms that cause disease? **(1)**

- ☒ **A** bacteria
- ☒ **B** fungi
- ☒ **C** pathogens
- ☒ **D** protozoa

(ii) Explain why it is not possible to treat Ebola using antibiotics. **(2)**

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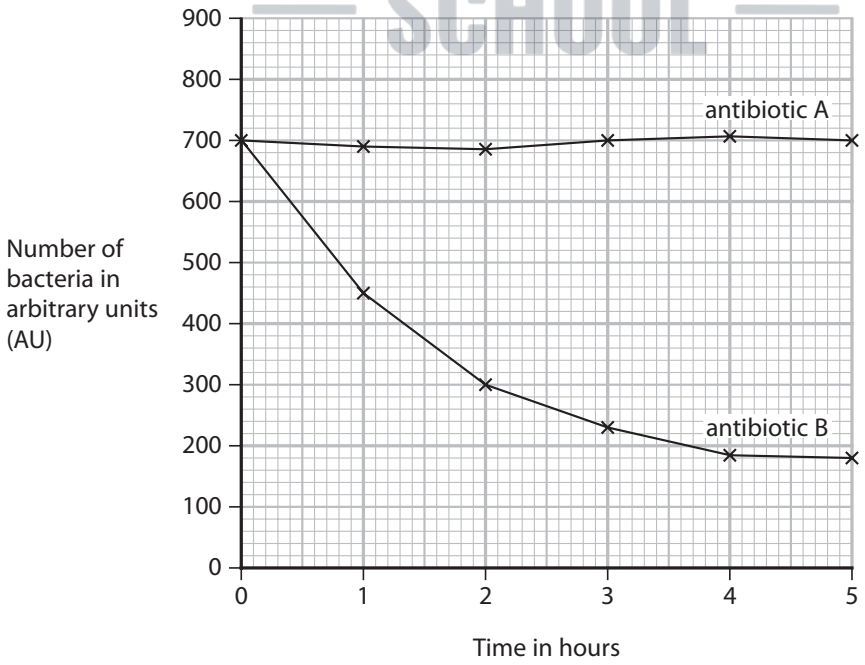
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(b) The graph shows the effect of two different antibiotics on the growth of a population of bacteria.



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2. Question 7 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

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(i) Calculate the mean rate of change of the bacterial population for antibiotic B during the five hours.

Give your answer to 2 significant figures.

(3)

mean rate of change = ... AU per hour

(ii) A bactericidal antibiotic kills bacteria.

A bacteriostatic antibiotic prevents bacteria from reproducing.

Explain which antibiotic is bactericidal.

Use data from the graph to help your answer.

(2)

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2. Question 7 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

(iii) Discuss which antibiotic would be better to use to treat any disease caused by the bacteria.

(3)

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(Total for Question 7 = 11 marks)

Working and answer

3. Question 7 · part b.iii · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

7 Various diseases are caused by microorganisms. Some diseases can be treated by using antibiotics.

(a) (i) Which of these is the name given to any microorganisms that cause disease? **(1)**

- ☒ **A** bacteria
- ☒ **B** fungi
- ☒ **C** pathogens
- ☒ **D** protozoa

(ii) Explain why it is not possible to treat Ebola using antibiotics. **(2)**

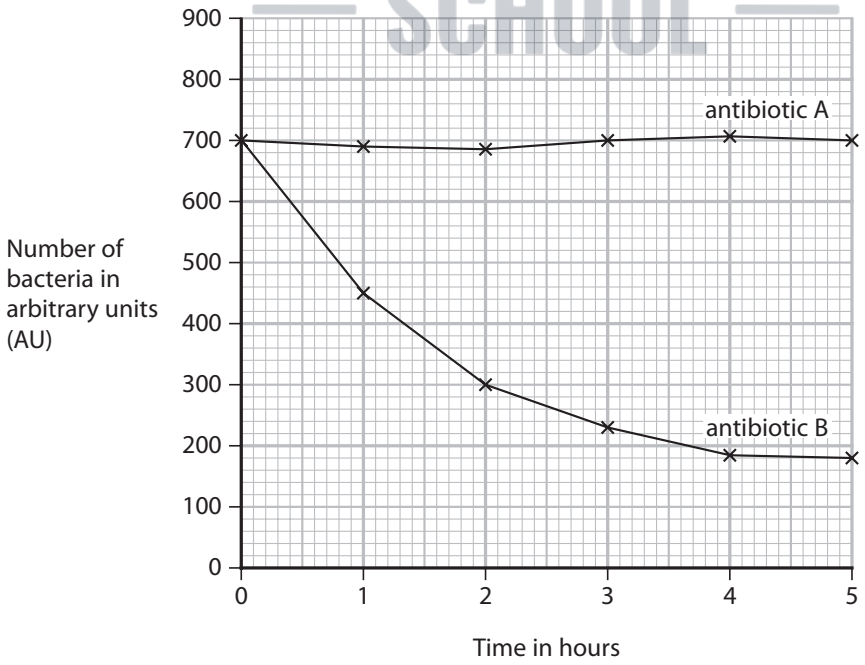
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(b) The graph shows the effect of two different antibiotics on the growth of a population of bacteria.



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3. Question 7 · part b.iii · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

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(i) Calculate the mean rate of change of the bacterial population for antibiotic B during the five hours.

Give your answer to 2 significant figures.

(3)

mean rate of change = ... AU per hour

(ii) A bactericidal antibiotic kills bacteria.

A bacteriostatic antibiotic prevents bacteria from reproducing.

Explain which antibiotic is bactericidal.

Use data from the graph to help your answer.

(2)

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3. Question 7 · part b.iii · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

(iii) Discuss which antibiotic would be better to use to treat any disease caused by the bacteria.

(3)

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(Total for Question 7 = 11 marks)

Working and answer

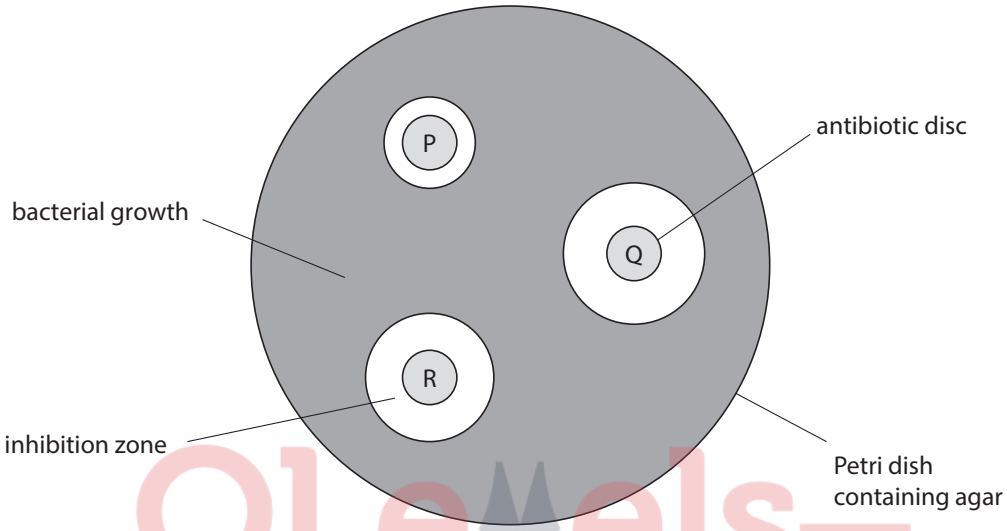
4. Question 5 · part a.iii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.iii only.

5 (a) A student investigates the action of three different antibiotics, P, Q and R, on the growth of bacteria.

The diagram shows the results of the investigation.



(i) Which is the independent variable in this investigation?

(1)

- ☒ **A** the mass of bacteria added
- ☒ **B** the size of the inhibition zone
- ☒ **C** the type of antibiotic added
- ☒ **D** the mass of agar in the Petri dish

(ii) Give two procedures that the student should use to reduce the risk of contamination by unwanted bacteria.

(2)

1 .....

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2 .....

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4. Question 5 · part a.iii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.iii only.

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(iii) Explain which is the most effective antibiotic.

(2)

(iv) The student concludes the bacteria are resistant to antibiotic P.

Explain how antibiotic resistance occurs in bacteria.

(4)

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(b) Decomposers are microorganisms such as bacteria and fungi that do not cause disease.

Describe the role of these microorganisms in the environment.

(2)

(Total for Question 5 = 11 marks)

4. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

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5. Question 7 · part b.ii · 4 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

7 Garlic plants contain a substance called allicin. This substance is thought to have antibacterial properties.

A student investigates the antibacterial effect of garlic.

This is the student's method.

- label three test tubes A, B and C
- add 10 cm^3 of milk into each of the tubes
- add a sample of the bacterium *E. coli* into tubes B and C using aseptic techniques
- crush the garlic in water to form garlic juice
- add 1 cm^3 of garlic juice to tube C
- shake all three tubes and leave to incubate for two hours
- place a sample from tube A into the centre of Petri dish A, from tube B into the centre of Petri dish B and from tube C into the centre of Petri dish C
- incubate the three Petri dishes for five days
- each day measure the diameter of any bacterial growth that has occurred

The table shows the student's results.

Petri dish	Diameter of bacterial growth in mm				
	day 1	day 2	day 3	day 4	day 5
A	0	1.7	3.0	4.6	7.1
B	0	4.2	8.4	15.1	36.5
C	0	0	0	0	0

(a) (i) State three control variables in the student's investigation.

(3)

1 .....

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2 .....

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3 .....

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5. Question 7 · part b.ii · 4 marks

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Further practice · Attempt part b.ii only.

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(ii) State the dependent variable in the student's investigation.

(1)

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(b) (i) Calculate the mean daily increase in the diameter of the bacterial growth for dish A and dish B.

(2)

mean daily increase for dish A ..... mm/day

mean daily increase for dish B ..... mm/day

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(ii) Explain the difference between the mean daily increases in the diameter of the bacterial growth for dish A and dish B.

(4)

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(c) Explain what is meant by an aseptic techniques for the transfer of bacteria.

(2)

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5. Question 7 · part b.ii · 4 marks · Working and answer

Working and answer

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